OPEN IMMERSION EQUALS UNIVERSALLY INJECTIVE ETALE EQUALS FLAT MONOMORPHISM OF FINITE PRESENTATION

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 02GP: sources/stacks-chow/tags.txt, source lines 3236–3236. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GR: sources/stacks-chow/tags.txt, source lines 3238–3238. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05VH: sources/stacks-chow/tags.txt, source lines 7551–7551. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06NC: sources/stacks-chow/tags.txt, source lines 8531–8531. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Theorem 0.1. *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent:*

(1) *f is an open immersion,*
(2) *f is universally injective and étale, and*
(3) *f is a flat monomorphism, locally of finite presentation.*

Proof. An open immersion is universally injective since any base change of an open immersion is an open immersion. Moreover, it is étale by Morphisms, Lemma 02GP. Hence (1) implies (2).

Assume f is universally injective and étale. Since f is étale it is flat and locally of finite presentation, see Morphisms, Lemmas 02GS and 02GR. By Lemma 05VH we see that f is a monomorphism. Hence (2) implies (3).

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Assume f is flat, locally of finite presentation, and a monomorphism. Then f is open, see Morphisms, Lemma 01UA. Thus we may replace Y by $f(X)$ and we may assume f is surjective. Then f is open and bijective hence a homeomorphism. Hence f is quasi-compact. Hence Descent, Lemma 06NC shows that f is an isomorphism and we win. $\square$